

Filtering image	• LH ↓, FSH ↓ • Hypogonadotropic hypogonadism • Low energy levels • ↓ libido • Erectile dysfunction
Heart	• Restrictive cardiomyopathy (not present in Wilson diseases)
Pancreas	• Type 1 DM (not seen in Wilson diseases)
Arthritis	• 2nd/3rd metacarpophalangeal joints involved → Ulnar deviation of fingers • Small joints of hands involved (Wilson diseases : Larger weight bearing joints gets affected)
Skin	• Slate/grey discoloration of skin due to excess iron getting deposited in tissues
Liver	• Cirrhosis • Portal hypertension

WORKUP

- Iron profile
 - Serum iron : rises
 - TIBC : low
 - Percentage saturation of Transferrin >45%
 - Serum ferritin >300 µg/L
 } Both needed for diagnosis
- MRI Liver (T2 imaging)
 - Quantify the amount of iron present in the liver
- IOC: **Liver biopsy with estimation** of iron content and determination of hepatocyte iron index
- HbA1c: Patients having Type I Diabetes mellitus
- Echocardiography: Restrictive cardiomyopathy
- X-ray joints: Degenerative changes in small joints of the hands (2nd/3rd metacarpophalangeal joints)

DETERMINATION	NORMAL	SYMPTOMATIC HEMOCHROMATOSIS
Plasma iron, µmol/L (µg/dl)	9-27	32-54
Total iron-binding capacity, µmol/L (µg/dl)	45-66	36-54
Transferrin saturation, %	22-45	50-100
Serum ferritin, µg/dl		1000-6000
Men	20-250	
Women	15-150	
Liver iron, µg/g dry weight	300-1400	6000-18000
Hepatic iron index (derived from liver iron)	<1.0	>2



• Plasma exchange for Hereditary Hemochromatosis (1 unit can remove about 250 mg of iron) → done twice/thrice

- Iron chelators for secondary hemochromatosis
 - **Deferoxamine**: Started when
 - Start when serum ferritin >1000 ng/ml (seen after 10-20 transfusions) → **S/C** deferoxamine used
 - Liver iron concentration measured by MRI (>3-5 mg/g dry weight)
- Oral drugs:
 - Deferasirox
 - Deferiprone

MCQs

00:45:59

Q. A 15-year-old child has presented to you with a history of dystonia and poor school grades. On examination, he is noticed to have slurred speech, tremors, abdominal distension with fluid thrill. Eye examination is shown below. Which is the initial investigation recommended for the patient?

- Slit lamp examination
- Serum copper
- 24-hour urinary copper
- Liver biopsy

Ans: **(c)**



ALCOHOLIC HEPATITIS



- Score to assess severity of alcoholic hepatitis: **Maddrey Discriminant function/score**
- DOC for alcoholic hepatitis: Prednisolone

ALCOHOLIC LIVER DISEASES

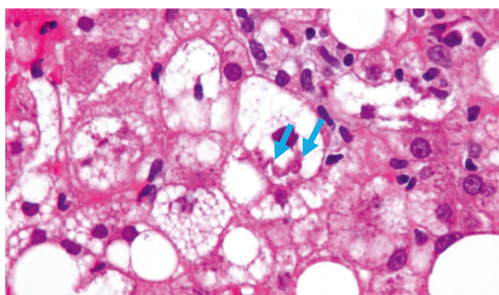
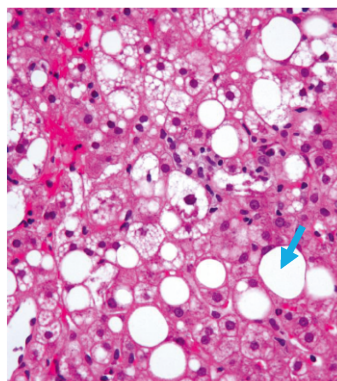
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- **3rd largest** contributor of disease burden
- Marker for heavy alcohol consumption: **GGTP** (Gamma glutamyl transpeptidase)
- Marker for alcoholic hepatitis : **SGOT/SGPT ratio >1**
 - SGPT is in the cytoplasm
 - Alcohol metabolism depletes B6 (pyridoxal phosphate) which is a cofactor required for SGPT synthesis
 - Thus denominator SGPT ↓
 - SGOT is in mitochondria
 - Alcohol majorly causes damage to mitochondria → SGOT ↑

SPECTRUM

00:04:20

- Fatty liver → alcoholic hepatitis → Cirrhosis
- Pathological findings of alcoholic hepatitis
 - Ballooning of hepatocytes
 - Polymorphonuclear cell infiltration
 - Spotty necrosis
 - Both macrovesicular > microvesicular fat deposition present→ Macrovesicular is the presentation of early alcoholic hepatitis
- **Mallory Denk bodies**
 - Formerly known as Mallory hyaline bodies
 - Cytokeratin in cytoplasm, **eosinophilic** in appearance



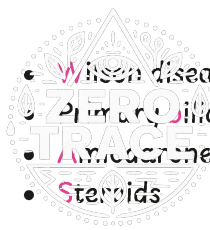
Mallory Denk bodies

CAUSES OF MALLORY DENK BODIES

00:06:19

(Mnemonic: **I** will **B**reak **A**lcohol **S**tatus)

- **Indian Childhood Cirrhosis**
 - Consumption of food made in brass utensils



- Wilson diseases
- Primary biliary cholangitis
- Amiodarone (treatment of ventricular tachycardia), Alcoholic hepatitis
- Steroids

ALCOHOL CONSUMPTION LEADING TO ALCOHOLIC LIVER DISEASES

00:08:08

MALE	FEMALE
• Threshold of damage to liver: 40-80 g/day → fatty liver • In eastern Europe and some parts of Russia: 160g/day	• Threshold of damage to liver: 20g/day → Alcohol liver diseases ○ Lower body mass index ○ Estrogen predisposes to liver damage

RISK FACTORS FOR ALCOHOLIC HEPATITIS

00:09:30

- Hepatitis C
- Obese
- MASH (Metabolic dysfunction associated steatohepatitis)/ earlier called as NASH
- Genetics

CLINICAL FEATURES

00:10:30

- Patient with h/o heavy alcohol intake
- Asymptomatic
- Right upper quadrant discomfort
- Nausea/Vomiting
- Jaundice – not seen initially
- Tender hepatomegaly
- History of heavy alcohol intake
- Late presentation:
 - Pulmonary erythema,
 - Spider nevi,
 - Temporal muscle vesting,
 - Hypersplenism,
 - Variceal bleed,
 - Caput medusae

DECOMPENSATED CIRRHOSIS	COMPENSATED CIRRHOSIS
• Variceal bleeding and caput medusae • Developing Portal hypertension	• No hypersplenism, variceal bleed, caput medusae • Spider nevi, palmar erythema, muscle wasting present

WORKUP

- Liver function test
 - SGOT/SGPT **> 1**
 - SGOT: **↑ 2-7 Times** above the baseline range
 - GGTP: Non-specific marker (just tells that person is chronic alcoholic)
- USG: Liver span ↑
 - Echotexture of liver (nodularity seen on cirrhosis)
- Doppler: **Portal vein flow reversal** + collaterals
- Coagulogram: PT/INR ↑



- **Fibroscan elastography** (Fibroscan): Normal value <6KPa
 - Higher values suggest progressive development of **stiffness** in the liver
- **Liver Biopsy:**
 - Helps to assess **severity and grade** the cirrhosis component

LIVER FIBROSIS STAGES (FO-F4)- HISTOPATHOLOGICAL EXAMINATION

STAGE	NAME	KEY FEATURE	MNEMONIC
F0	No Fibrosis	Normal liver tissue	Free of scars
F1	Portal fibrosis	Scarring starts around portal tract	P ortal
F2	Peripheral fibrosis	Scarring spreads beyond portal area	P eri-portal
F3	Bridging fibrosis	Scars from bridges between veins	B ridges
F4	Cirrhosis	Nodules form, liver shrinks	C irrhosis

SCORE TO ASSESS THE SEVERITY OF ALCOHOLIC HEPATITIS

- Maddrey Discriminant Function/ Score
 - $4.6 \times (\uparrow \text{PT above the average of normal limits}) + \text{serum bilirubin} > 32$
 - Maddrey score >32: Indication to start Prednisolone in patient
- MELD score >20
 - S. **C**reatinine
 - S. **B**ilirubin
 - **I**NR

MANAGEMENT

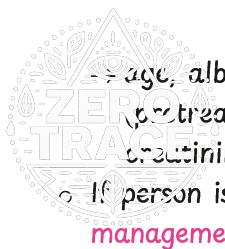
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- Alcohol De Addiction: **Naltrexone/Acamprosate**

DRUG	MOA
Naltrexone	• Block μ receptors in brain $\rightarrow$ inhibition of dopamine release $\rightarrow$ alcohol consumption $\downarrow$ • μ receptors $\rightarrow$ release dopamine $\rightarrow$ pleasure related to alcohol intake • Cuts off the craving component
Acamprosate	• To maintain the abstinence • Restores GABA/Glutamate balance in post-acute withdrawal

Important Information

- Disulfiram used for **aversion** in individuals having high levels of self-motivation
- DOC for alcoholic hepatitis: Prednisolone 32 mg X 4 weeks
- To predict steroid responsiveness
 - Lillie score value <0.45



-> age, albumin, bilirubin on day 0 or day 7 (pre-treatment and post-treatment values), creatinine and PT

- If person is not steroid responsive → **conservative management** and if possible → **early liver transplantation**

LILLIE SCORE	STEROID RESPONSIVENESS
<0.45	YES
>0.45	NO

- TNF-alpha inhibition: Pentoxifylline
- Check eligibility for early liver transplantation (select centers: MELD>25 and poor Lillie score - >0.45)

LABORATORY DIAGNOSIS OF ALCOHOLIC FATTY LIVER AND ALCOHOLIC HEPATITIS

TEST	COMMENT
AST	↑ two to seven-fold, <400 IU/L, greater than ALT
ALT	↑ two to seven-fold, <400 IU/L
AST/ALT	Usually >1
GGTP	Not specific to alcohol, easily inducible, elevated in all forms fatty liver
Bilirubin	May markedly ↑ alcoholic hepatitis despite modest elevation in alkaline phosphatase

NON-INVASIVE METHODS OF ASSESSING HEPATIC FIBROSIS AND CIRRHOSIS

METHODS	PARAMETERS	ADVANCED FIBROSIS	CIRRHOSIS
FIB-4	Age, AST, ALT, Platelet count	>1.45	>3.25
TE (Transient elastography)	Measures the speed of a shear wave generated by vibration through liver tissue	>7.3kPa	>15kPa (9-26.5 kPa)
ARFI (Acoustic Radiation Force Imaging)	Measures the speed of shear wave generated by acoustic radiation force through liver	>1.3 m/s	>1.87 m/s

METABOLIC DYSFUNCTION ASSOCIATED WITH STEATOHEPATITIS

00:24:14

- New drug approved by FDA: **Resmetirom**
 - MOA: Thyroid hormone receptor beta agonist that enhances lipid metabolism and reduces hepatic fat accumulation

COMPARISON TABLE OF EMERGING DRUGS IN MASH

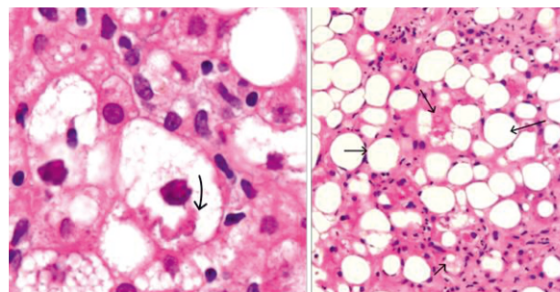
DRUG NAME	MECHANISM	KEY TRIAL FINDINGS	INDICATION STATUS	UNIQUE BENEFIT
Resmetirom	THR- β agonist (targets liver fat metabolism)	↓ liver fat, improved fibrosis in F2-F3 patients (MAESTRO-NASH trial)	FDA approved (March 2024)	First approved drug for MASH with fibrosis
Tirzepatide	Dual GLP-1 & GIP receptor agonist	52% MASH resolution (phase 2), ↓ weight, ↓ liver enzymes	Under study for MASH	Strong weight loss + liver protection
Semaglutide	GLP-1 receptor agonist	63% MASH resolution but less effect on fibrosis (ESSENCE trial)	Off-label use, under trial	Effective MASH reversal + weight loss
Pegzofermin	Lon-acting FGF21 analog	↓ Liver fat & fibrosis biomarkers (ENLIVEN trial)	Phase 2 complete Phase 3 ongoing	Weekly injection promising for fibrosis
Pioglitazone	PPAR- γ agonist	Improves histology in diabetic MASH patients	Off-label use	Best in diabetic MASH
Vitamin E	Antioxidant	↓ in steatohepatitis in non-diabetics with biopsy-proven MASH	Recommended for non-diabetics	Inexpensive oral option

- Vitamin E: ↑ cardiovascular mortality
- 1st line intervention for management is **weight loss and diet modification**
- M/c cause of liver transplantation in the world: MASH

MCQ's

Q. The liver biopsy specimen of a deceased patient with chronic alcoholism is being presented at a pathology conference. The patient passed away during an episode of binge drinking. Mark the option with the correct statements.

- Discrimination score is used to assess severity of AH
- The marker for heavy alcohol consumption is ALT
- Drug of choice for AH is Prednisolone
- Vitamin E reduces severity of AH



- b, d
- a, c
- a, d
- a, c, d

Ans. (b)



13. HEREDITARY HYPERBILIRUBINEMIA

CASE BASED SCENARIO

00:00:47

- A 24-year-old woman develops jaundice after a religious pilgrimage. LFT shows S. Bilirubin 5 mg/dL, unconjugated bilirubin 4.3 mg/dL, SGOT, SGPT, and SAP are normal. She says her mother remarked that she has had yellow eyes since childhood, and she has shown multiple doctors in the last 3 years. Viral markers are negative. No history of drug intake or alcohol consumption
 - Diagnosis: **Gilbert syndrome**
 - M/c variety of hereditary hyperbilirubinemias
 - Defective conjugation as the activity of **UDP-Glucuronyl transferase** enzyme in the liver is ↓
 - Icterus appears/ sclera turns yellow: >3 mg/dL
 - Jaundice here will have **waxing and waning** course
 - **Cause:** Deficiency of UDPGT activity due to defect in UGT1A1 gene
 - Encodes for UDPGT enzyme activity
 - In majority, activity of enzyme is <50%
 - **Triggers:** Fasting and exertion, illness, stress, hormone changes like pregnancy

UGT1A1 ACTIVITY	CONDITION	BILIRUBIN LEVELS
~100%	Normal	<1 mg/dL
~30-50%	Gilbert syndrome (Gentle)	2-6 mg/dL
~10-30%	Crigler Najjar type 2 (Critical)	6-20 mg/dL
<10% (or none)	Crigler Najjar type 1	>20-45 mg/dL

- Crigler-Najjar can cause kernicterus/brain damage in newborns
- Gilbert syndrome has a **milder course** of presentation

GILBERT SYNDROME

00:06:04

- No definitive treatment required
- Mostly harmless, **benign** condition
- Pattern of inheritance
 - Promoter mutation UGT1A1 gene: **Autosomal Recessive** (M/c)
 - Missense mutation UGT1A1 gene: **Autosomal dominant**
- UDPGT activity ↓
- No role of Phenobarbitone
 - More beneficial in Crigler-Najjar syndrome type 1 as compared to Gilbert syndrome

CRIGLER NAJJAR SYNDROME

00:07:06

- UDPGT activity is reduced, and impaired conjugation of bilirubin
- **Type I:** **More Intense** and has immediate danger- UDPGT activity < 10%



- Neonatal presentation of pathological jaundice
- **Crigler-Najjar** - UDPGT activity 10-30% of normal
- Manifestations in the later part of childhood or adult age group

CASE BASED SCENARIO

- Neonate develops pathological jaundice on day 1 of life and is managed with phototherapy followed by exchange transfusion. Day 3 of life, the neonate develops a high-pitched cry with opisthotonus & seizures. S. Bilirubin post exchange transfusion again rises to 30 mg/dl. Diagnosis is?
 - **Opisthotonus**: Only the occiput and heel of the patient touch the bed
 - Other causes: Neonatal tetanus/day 8th illness, Strychnine poisoning
 - Diagnosis: **Crigler-Najjar syndrome type 1**
 - ↑ conjugated bilirubin → cross blood-brain barrier → damage to putamen
 - Bad prognosis from neurological perspective

Important Information

- Most common cause of pathological jaundice: **Rh incompatibility**
 - Routine Rh incompatibility-induced pathological jaundice responds to exchange transfusion
- Most common cause of kernicterus: **Crigler-Najjar syndrome**
 - Basal ganglia defective in kernicterus: Putamen
 - Neurological sequelae of Kernicterus:
 - **Athetoid cerebral palsy**
 - Hearing loss
 - Intellectual disability

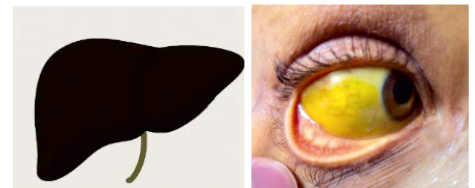
DUBIN JOHNSON SYNDROME

00:12:17

- Gene defect: **ABCC2** (ATP binding cassette subfamily 2)
- Protein defect: **MRP-2** (Multidrug Resistant Protein 2) - **Canalicular protein**

PATHOGENESIS

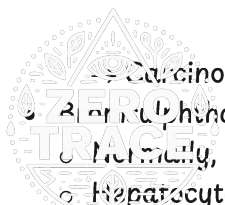
- **Conjugated bilirubin** not exported out of the hepatocytes → regurgitate into blood stream → appears in urine
- Manifestations:
 - Bile salt/acid handling is normal, so **pruritus is not seen**
 - **Bilirubinaemia** causing **mustard yellow discoloration** of urine
 - Icterus developing after triggers like OCP, pregnancy
 - Epinephrine-based pigments in hepatocytes → **Black liver**



00:15:28

WORK UP

- LFT:
 - S. Bilirubin: Conjugated value > **15%** of Total serum bilirubin
 - SGOT, SGPT, SAP, 5' Nucleotidase: normal
- MRCP imaging: Normal
 - To rule out extrahepatic causes of obstructive jaundice
 - Cholelithiasis
 - Strictures in CBD



- $\rightarrow$ Carcinoma pancreas
- Bromsulphalein test (BSP):
 - Normally, dye is cleared (in 60-90 minutes) from bloodstream as hepatocytes are able to take it up
 - Hepatocyte uptake is defective in Dubin-Johnson syndrome $\rightarrow$ Dye remains in bloodstream
- Urine coproporphyrin level is normal, but urine CP I > III levels (flipped ratio of urinary Coproporphyrin III and I)
 - In a normal person, Urinary CP III > I
 - $\rightarrow$ CP I is excreted in bile, and CP III in urine
 - In Dubin-Johnson syndrome, CP I is not excreted
 - $\rightarrow$ Regurgitates in blood stream $\rightarrow$ Appear in urine
- Oral cystography: Gall bladder is not visualized (poor flow of bile)
- Genetic studies (IOC) and Liver biopsy showing pigmented black hepatocytes (epinephrine-derived)

ROTOR SYNDROME

00:21:00

- Autosomal Recessive
- Genes involved: SLCO1B1 and SLCO1B3
- Encoded proteins that are defective: OATP1B1 (organic anion transporting polypeptides)

PATHOLOGY

- OATP1B1 is an import pump and cannot take up coproporphyrins from blood into hepatocytes
 - So all of it gets dumped from blood into urine
- Gall bladder visualized on oral cystography
- Total urinary coproporphyrin levels: $\uparrow$
- Coproporphyrin III / I levels: $\uparrow$

	DJS	ROTOR SYNDROME
Total urinary coproporphyrin	Normal	Increased
Liver colour	Black	Normal
Main issue with hepatocytes	Can not export into bile due to defect of MRP 2	Can not take in from coproporphyrins blood due to defect of OATP



Q. During exploratory laparotomy, the surgeon made a note of the black pigmentation of the liver of a patient who had presented with features of intestinal obstruction. He has a past history of multiple episodes of jaundice that used to occur after beer intake. Which genetic defect is likely to cause this presentation?

- a. UGT1A1
- b. SLCO1B1
- c. ATP7B
- d. ATP7A

Ans. (b)

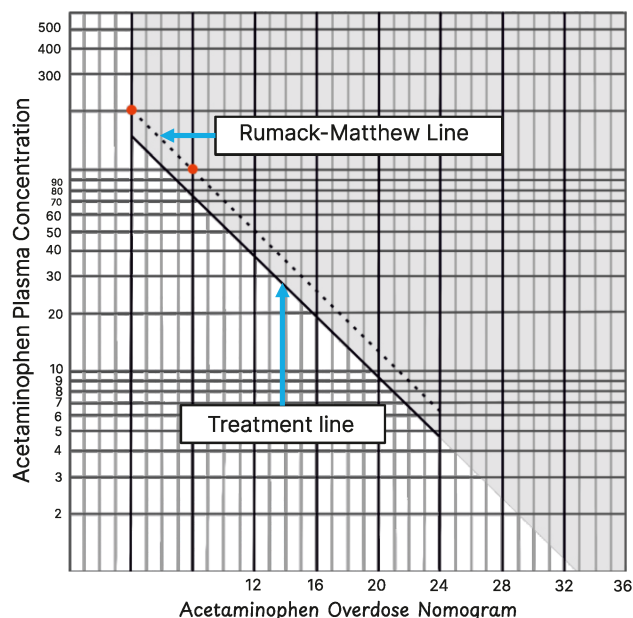


PARACETAMOL AND IRON TOXICITY

PARACETAMOL TOXICITY

00:00:28

- Nomogram helps to decide the threshold or critical level beyond which intervention in form of N-acetyl cysteine is done
- **Rumack-Matthew Line:** 200 $\mu\text{g/ml}$ of serum paracetamol levels at 4 hours or $>100 \mu\text{g/ml}$ at 8 hours $\rightarrow$ Values beyond which N-acetylcysteine should be given and less than these values conservative management is done
- **Treatment line:** Drawn to left of main Rumack-Matthew Line
 - Threshold is 25% lower
- Safe dose of PCM in a normal adult : **$<3 \text{ g/day}$**
- Safe dose of PCM in an alcoholic adult : **$<2 \text{ g/day}$**
- If PCM + Opioids prescribed, FDA limits the amount of PCM to **325 mg**
- Fatal dose **$>25\text{g/day}$**
- Cut off at which N-acetyl cysteine should be administered: N-acetyl cysteine helps in regeneration of Glutathione $\rightarrow$ Counterbalance the harmful metabolites of Paracetamol
 - $>200 \mu\text{g/ml}$ at 4 hours of ingestion
 - $100 \mu\text{g/ml}$ at 8 hours of ingestion



METABOLISM OF PCM

00:04:31

PHASE 1

- Cytochrome P450 produces hepatotoxic product **N-acetyl p-benzoquinone imine (NAPQI)**
- This is neutralized by **Glutathione**
- Excess PCM depletes Glutathione levels $\rightarrow$ NAPQI $\rightarrow$ Hepatotoxic $\rightarrow$ Damage to the liver
- Low glutathione levels are seen in alcoholic patients, starvation, Isoniazid, Barbiturates
- NAC contains **Sulphydryl groups** that helps in Glutathione regeneration $\rightarrow$ Neutralisation of NAPQI

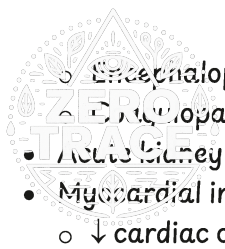
PHASE 2

- Conjugation (major metabolism into harmless product)
 - Sulfate moiety
 - Glucuronidation

CLINICAL FEATURES OF PCM OVERDOSE

00:06:51

- At **4-12 hours:** Nausea, vomiting, diarrhoea, abdominal pain, and shock
- At **24-48 hours:** Fulminant hepatic failure $\rightarrow$ Manifestations of fulminant hepatic failure d/t $\uparrow$ conc. of ammonia



- Encephalopathy (prior to this asterix can be seen)
- Encephalopathy (synthesis of clotting factors in liver)
- Acute kidney injury: Anuria
- Myocardial injury:
 - ↓ cardiac output
 - Can worsen the acute kidney injury
 - Development of metabolic acidosis leading to cardiogenic shock

TREATMENT

- Gastric lavage
 - Cholestyramine
 - N-acetylcysteine → Start ideally within <8 hour of ingestion upto 24-36 hours
 - It can be given orally or intravenously (20 doses given)
 - Orthoptic liver transplant indicated if signs of liver failure despite medical treatment and rising serum lactate values >3.5 mmol/L
- Not effective after 30 minutes of ingestion

KING'S CRITERIA

(mnemonic: **PHC**)

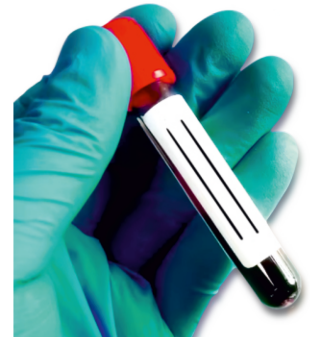
pH <7.3 in spite of fluid resuscitation irrespective of encephalopathy grade

All 3 should be present

- PT >100 secs or INR >6.5
- Hepatic encephalopathy grade III or grade IV
- S. Creatinine >3.4 mg/dl

SAMPLING FOR SERUM PARACETAMOL

- Test required: Serum paracetamol levels
- Clinical details: 22-year-old female, suspected paracetamol overdose, ingestion approximately 5 hours ago
- Action: **STAT sample**- emergency processing requested
- Purpose: For interpretation with Rumack-Mathew nomogram to guide NAC therapy



00:11:59

IRON TOXICITY

- Leading cause of poisoning in children <6 years of age

STAGES OF IRON TOXICITY

→ Stage 1

- GI features: Nausea, vomiting, diarrhoea and GI haemorrhage

→ Stage 2 (latent phase)

- Deceptive phase as vomiting stops

→ Stage 3

- Metabolic acidosis and cardiac damage: highest chances of death



Stage 4

• Coagulopathy and Encephalopathy (Fulminant hepatic failure)

Stage 5 (delayed phase)

- Scarring of gut with progression of Cirrhosis

MANAGEMENT

- IV fluids → Hypovolemic shock and GI bleeding
- Oxygen supplementation
- No role of activated charcoal and ipecac → Worsens the GI haemorrhage
- Antidote: **Deferoxamine (IV)** → Chelates iron and used when
 - Values **>350 µg/dl** with features of toxicity
 - Values of iron **> 500 µg/dl** without toxicity features

FEATURES OF TOXICITY

- Metabolic acidosis
- Altered mental status
- Protracted vomiting



- Q. A 5-year-old child is brought with complaints of ingesting a large dose of paracetamol syrup. She is confused and is vomiting constantly. Serum PCM levels are $150 \mu\text{g/ml}$ at 6 hours of consumption. Which is not correct about this patient?
- a. Plot value of Rumack Mathew nomogram
 - b. N-acetylcysteine infusion
 - c. Start activated charcoal
 - d. Run check for King's criteria

Ans. (c)

- Q. A 3-year-old child is brought to the emergency department 4 hours after accidentally ingesting several Ferrous Sulfate tablets. He initially had vomiting and diarrhoea but now appears stable. An abdominal x-ray reveals multiple radio-opaque tablets in the stomach. Which of the following is the most appropriate next step in management?
- a. Discharge the child as symptoms have resolved
 - b. Start IV deferoxamine immediately
 - c. Give activated charcoal to reduce iron absorption
 - d. Admit and observe closely for impending cardiac depression and systemic toxicity

Ans. (d)

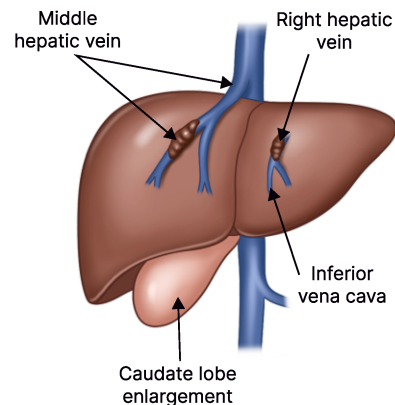


BUDD CHIARI SYNDROME



TRIAD FOR MANIFESTATION OF BUDD-CHIARI SYNDROME

- **Abdominal pain** (right upper quadrant):
 - Due to stretching of Glisson's capsule of liver, (have nerve endings)
- **Ascites** (fast onset)
 - Due to ↑ congestion in hepatic sinusoids → gross abdominal distension
- **Hepatomegaly**
- **Acute onset** presentation can lead to death of the individual if undiagnosed



00:01:50

DEFINITION

- Thrombosis of **≥2 hepatic veins**
- USG may show **Caudate lobe enlargement** → may or may not be present
- **Post-hepatic** cause of portal hypertension
 - Congestion in liver → pressure in portal vein ↑

CAUSES

00:02:55

HEMATOLOGICAL DISORDERS (M/c)	• Polycythemia vera: JAK 2 mutation ◦ Uncontrolled proliferation of normoblasts → ↑ RBC count → sluggish flow of blood • Paroxysmal nocturnal hemoglobinuria: Defect of PIGA gene ◦ Missing CD-59 → platelets show spontaneous aggregations → clot formation in hepatic veins → manifestations of budd chiari syndrome
DEEP VEIN THROMBOSIS	• Hypercoagulable states : Factor V Leiden mutation → unchecked activity of factor V → activated clotting system
POSTPARTUM, COMBINED ORAL CONTRACEPTIVE PILLS WITH ↑ ESTROGEN	
TUMORS	• Tumor pressing on hepatocytes from outside → sluggish flow of blood through hepatic veins • Hydatid cyst of ↑ size can compress hepatic veins → Manifestations of Budd-Chiari syndrome • Tumors can be: ◦ HCC ◦ Metastasis



INFECTIONS	Hydatid cyst
ACQUISITIVE TISSUE DISORDERS	Anti-phospholipid antibody syndrome - Anti-beta2 glycoprotein antibody → triggering of the intrinsic system of clotting Behcet's disease - Vasculitis involving arteries and veins → Cerebral venous thrombosis and venous stroke can occur Sarcoidosis - Non-caseating granuloma → press on hepatic veins → Budd chiari syndrome
MEMBRANOUS DEFECTS IN IVC AND PORTAL VEINS	
TOTAL PARENTERAL NUTRITION	IVC cannulation

CLINICAL FEATURES

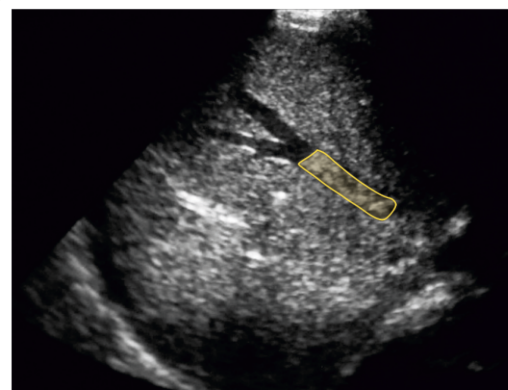
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- Triad:**
 - Abdominal pain
 - Ascites
 - Hepatomegaly
- Jaundice
- Hepatorenal syndrome : **Anuria**
- Pedal edema: Venous return to right side of heart is impeded due to occlusion of hepatic veins
- Caput medusae if chronic presentation
- If all 3 hepatic veins are involved → venous return to heart compromised → death

WORKUP

00:10:20

- USG with Doppler** (initial test done): Caudate lobe enlargement may or may not be present
 - On Doppler: Single drop occurring
- Ascitic tap
 - Serum Albumin Ascitic Gradient (**SAAG**) **>1.1 g/dL**
 - Ascitic fluid protein **> 2.5 g/dL (early BCS)** → transudative ascitic fluid accumulation
 - Late BCS: Ascitic fluid protein < 2.5 gm/dL → lesser movement of proteins due to fibrosis
- LFT: Serum bilirubin ↑
- MRI abdomen
- IOC : **Hepatic venography**



USG WITH DOPPLER



ACUTE	CHRONIC
EASL guidelines • Balloon angioplasty • Anticoagulation	• T.I.P.S • Spironolactone • End Stage Liver Diseases → Orthoptic Liver transplantation

- $SAAG \approx 0.012 \times HVP + 0.43$
 - HVP - Hepatic Venous Pressure Gradient
 - If HVP $\uparrow$, SAAG will also $\uparrow$



Q. A 36-year-old woman presents with progressive abdominal distension, right upper quadrant pain, and pedal edema. On examination, there is tender hepatomegaly and tense ascites. A Doppler ultrasound of the liver demonstrates absent flow in the hepatic veins. Past medical history is that she was on steroids for sarcoidosis. Which test will be done next to confirm the diagnosis of this condition?

- a. MRI Liver
- b. Hepatic venography
- c. Contrast-enhanced CT scan
- d. Ascitic tap

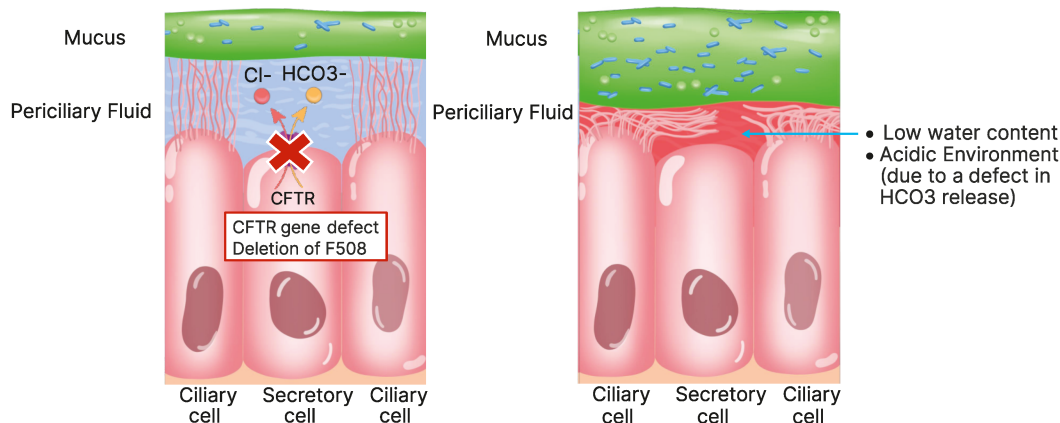
Ans. (b)

Q. Which of the following is the most common underlying cause of Budd-Chiari syndrome?

- a. Oral contraceptive use
- b. Paroxysmal nocturnal haemoglobinuria
- c. Antiphospholipid antibody syndrome
- d. Myeloproliferative disorders

Ans. (d)

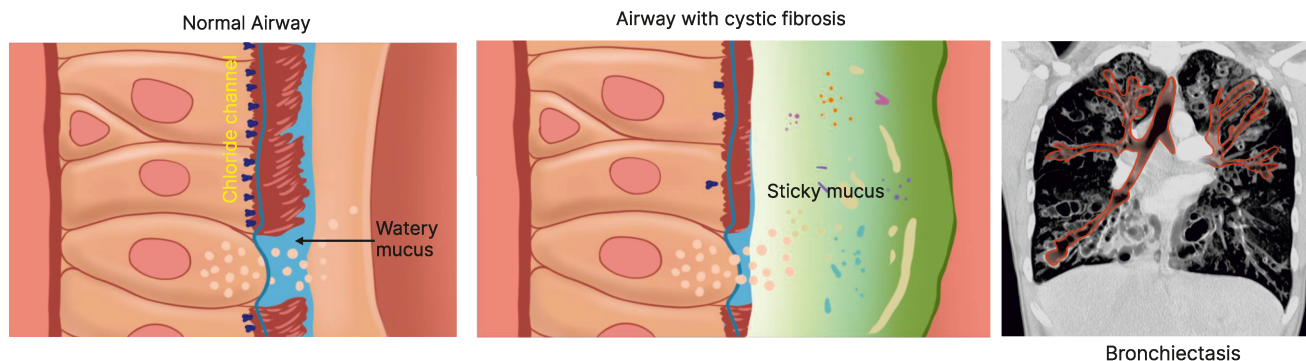
- Why recurrent Pneumonia occurs in Cystic fibrosis patients?



- Low water content of PCL (Periciliary fluid) : D/t CFTR gene defect & F508 mutation → Secretions become viscid
- Acidic Environment of Periciliary fluid due to a defect in HCO₃ release
- Hampered mucociliary clearance → Stasis of secretions → Infection
- Mortality in Cystic fibrosis patients is due to lung infections

PATHOGENESIS OF CYSTIC FIBROSIS

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- Lungs: Bronchiolitis and bronchitis → Recurrent pneumonia → Bronchiectasis
- Progressive lung damage → Respiratory failure
- Inheritance pattern: Autosomal recessive
- Defect in chromosome: **chromosome 7**
- M/c mutation: **Delta F508 mutation**
 - Defective/misfolded CFTR protein due to deletion of amino acid phenylalanine at 508 position
- Normal **CFTR protein** has 1480 aminoacids
 - Assists in Cl⁻, HCO₃⁻, H₂O transport
- Aka Mucoviscidosis/Exocrinopathy